Growth Chamber



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Growing chamber

What it is used for

A growth chamber provides controlled environmental conditions such as temperature, humidity, and lighting for growing plants or conducting laboratory studies. Use the chamber according to the approved experimental protocol and the manufacturer's instructions.

Main parts

- Insulated chamber and access door
- Temperature control system and sensors
- Humidity control system, if fitted
- Lighting system and timer
- Shelves, trays, or sample racks
- Control panel, display, and power connection

Before you start

1. Confirm the chamber is suitable for the intended plant, sample, or experiment.
2. Inspect the chamber, door seals, shelves, sensors, and power cord.
3. Clean the interior and shelves using an approved method.
4. Check calibration or verification status of temperature and humidity sensors.
5. Confirm the required temperature, humidity, lighting, and timing settings.
6. Ensure ventilation openings are clear and samples are correctly labelled.

How to use

1. Place the chamber in a suitable location with the required clearance for ventilation.
2. Connect the unit to the recommended power supply.
3. Switch on the chamber and allow it to stabilise before loading samples.
4. Set temperature, humidity, lighting, and timer according to the approved protocol.
5. Verify displayed conditions using the available monitoring system.
6. Arrange labelled samples on shelves without blocking airflow or lighting.
7. Close the door securely and confirm the chamber maintains the required conditions.
8. Monitor readings and record environmental conditions at the required intervals.
9. Respond to alarms or deviations according to the laboratory procedure.
10. Avoid unnecessary door opening during the growth or test period.
11. At completion, remove samples using appropriate handling and PPE.
12. Switch off the chamber if required by the protocol and clean the interior.
13. Record the run details, deviations, and maintenance needs.

⚠ Safety

- Only trained personnel should configure and operate the chamber.
- Follow laboratory procedures for biological, chemical, or plant materials stored inside.
- Do not block ventilation paths or overload shelves.
- Avoid touching hot, cold, or electrical components without appropriate protection.
- Do not bypass alarms, interlocks, or temperature protections.
- Disconnect power before internal cleaning or maintenance when required by the manufacturer.

Care & storage

- Clean spills promptly using compatible cleaning products.
- Inspect door seals, sensors, lighting, and ventilation openings regularly.
- Keep a record of calibration, temperature checks, and maintenance.
- Arrange servicing if readings drift, alarms persist, or controls malfunction.

Common mistakes to avoid

- Setting incorrect environmental conditions for the experiment.
- Overloading shelves and restricting airflow.
- Ignoring alarms or failing to record environmental readings.
- Cleaning with chemicals that damage chamber surfaces or sensors.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.